

Everything SATS — Maths Mock 3

Paper 3: reasoning

This is an original practice paper written for Everything SATS. It follows the layout and question types of the real Key Stage 2 mathematics test, but it is **not** a real past paper.

21 questions, 35 marks. Some questions ask you to show your method. You may get a mark for a correct method even if the answer is wrong. You may not use a calculator.

In the real test this paper lasts 40 minutes. There is no need to time yourself here — take the time you need.

For the adult: the answers and how to award marks are in the separate **answer booklet** (Maths Mock 3 answers).

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Name: _____

Paper 3: reasoning

1

Tick **all** the numbers that are less than one million two hundred thousand.

1,204,000

1,199,000

1,190,100

1,500,000

1 mark

Circle the numbers that have 6 in the **thousands** place.

830,922

536,111

895,175

135,344

536,396

1 mark

2

There were 92,905 people at a football match.

Round this number to the nearest 100.

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1 mark

3

Millie buys 2 kilograms of apples and spends £2.42.

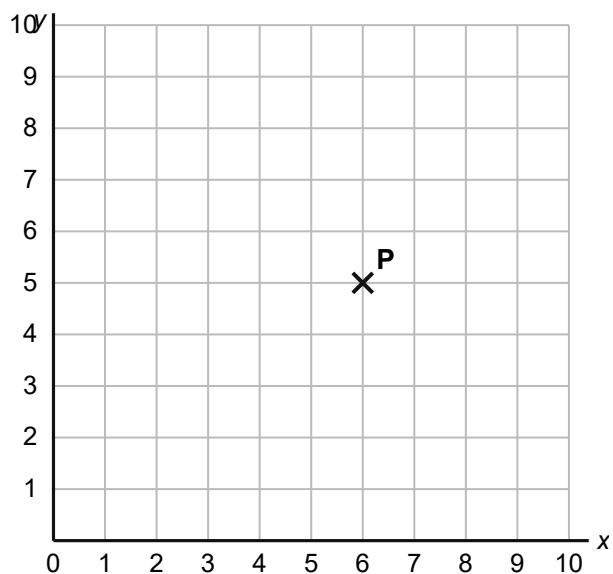
What is the cost of one kilogram of apples?

£	
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1 mark

4

Point P is at (6, 5) on the grid.



Point P is translated 4 squares right, 4 squares down and then 1 square left. What are the coordinates of point P after the translations?

(,)

1 mark

5

The mass of an empty van is 3,000 kilograms. It is loaded with 7 crates. Each crate has a mass of 75 kilograms.

What is the total mass of the van and its load?

Show your method

kg

2 marks

6

Draw lines to match each improper fraction to the equal mixed number.

$$\frac{19}{4}$$

$$6\frac{3}{4}$$

$$\frac{27}{4}$$

$$4\frac{3}{4}$$

$$\frac{17}{4}$$

$$7\frac{3}{4}$$

$$\frac{23}{4}$$

$$5\frac{3}{4}$$

$$4\frac{1}{4}$$

2 marks

7

William buys these three items.

cereal
£2.32

milk
£1.09

bread
£1.10

William pays with a £10 note. How much change does William get?

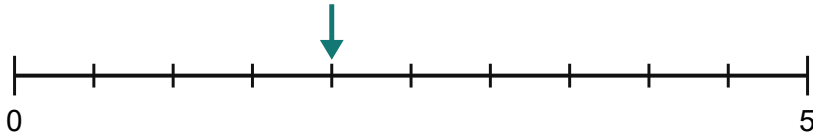
Show your method

£

2 marks

8

Here is part of a number line.



Write the number that the arrow is pointing to.

1 mark

Write the number that is 12 less than 5.

1 mark

9

Here is some information about four animals in a zoo.

	Elephant	Hippo	Giraffe	Rhino
Mass	6,000 kg	1,400 kg	1,300 kg	2,000 kg
Height	3.3 m	1.6 m	5 m	1.8 m

The giraffe is more than three times as tall as the hippo.

The rhino is 20 cm taller than the hippo.

The elephant is more than twice as heavy as the rhino.

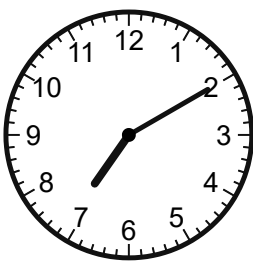
The hippo and the rhino together are heavier than the elephant.

Tick the statements that are true.

2 marks

10

This clock shows a time in the afternoon or evening (pm).



Write the time using the 24-hour clock.

1 mark

11

0.78 $\frac{9}{100}$ 0.782 0.98

Write these numbers in order, starting with the **smallest**.

1 mark

12 Look at this expression: **$y + 4$**

Tick the value of y that makes $y + 4$ a **prime number**.

$y = 8$	☐
$y = 6$	☐
$y = 7$	☐

1 mark

Tick the value of y that makes $y + 4$ a **square number**.

$y = 3$	☐
$y = 18$	☐
$y = 21$	☐

1 mark

13 A factory packs pencils and rubbers into boxes.

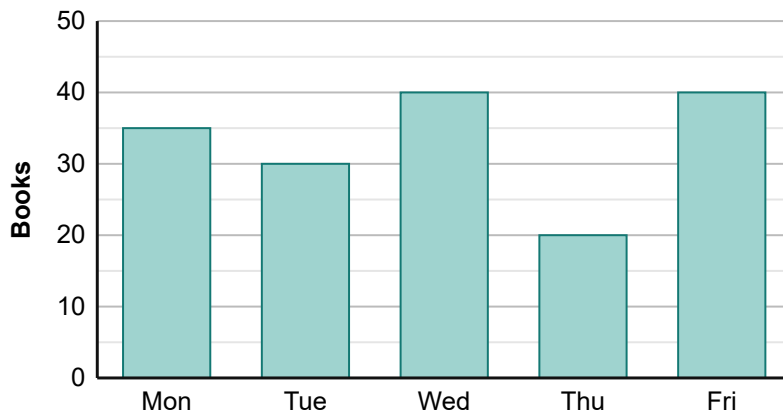
There are 24 pencils in a box. How many pencils are there in 100 boxes?

1 mark

There are 25 rubbers in a box. How many boxes are needed for 2,700 rubbers?

1 mark

14 This bar chart shows how many books were borrowed from a library each day.



How many more books were borrowed on Wednesday than on Thursday?

1 mark

15

A shop buys 52 boxes of juice cartons. Each box holds 48 cartons. On average, the shop sells 64 cartons each day.

How many days will it take for all of the cartons to be sold?

Show your method

days

3 marks

16

Convert 8.12 km to metres.

m

1 mark

Convert 33.3 cm to metres.

m

1 mark

17

Priya has some wooden blocks that are all the same size, and builds two towers. The taller tower has 9 blocks and is 13.5 cm tall.

The smaller tower has 4 blocks. What is the height of the smaller tower?

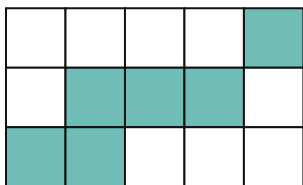
Show your method

cm

2 marks

18

This shape is made from equal squares.



What fraction of the shape is shaded? Give your answer in its simplest form.

1 mark

19

Here is some information about a number:

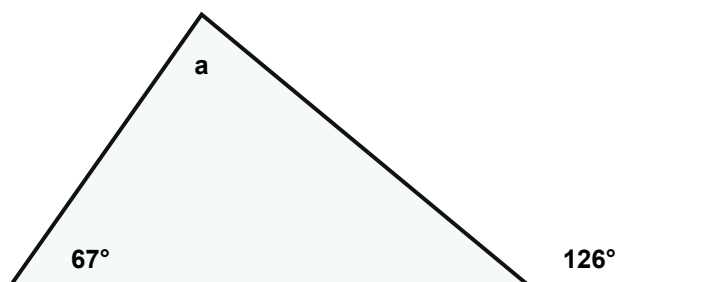
- It has two digits.
- It is a multiple of 7.
- One of its digits is 7.

Write **all** the possible numbers it could be.

2 marks

20

This triangle has one side extended to make a straight line.



Not to scale

Calculate the size of angle a .

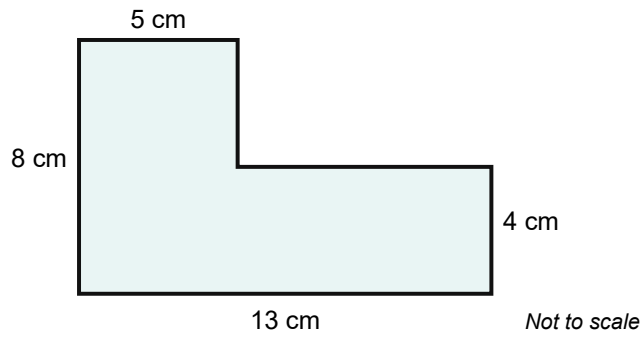
Show your method

°

2 marks

21

Here is a shape made from rectangles.



cm²

What is the area of the shape?

1 mark